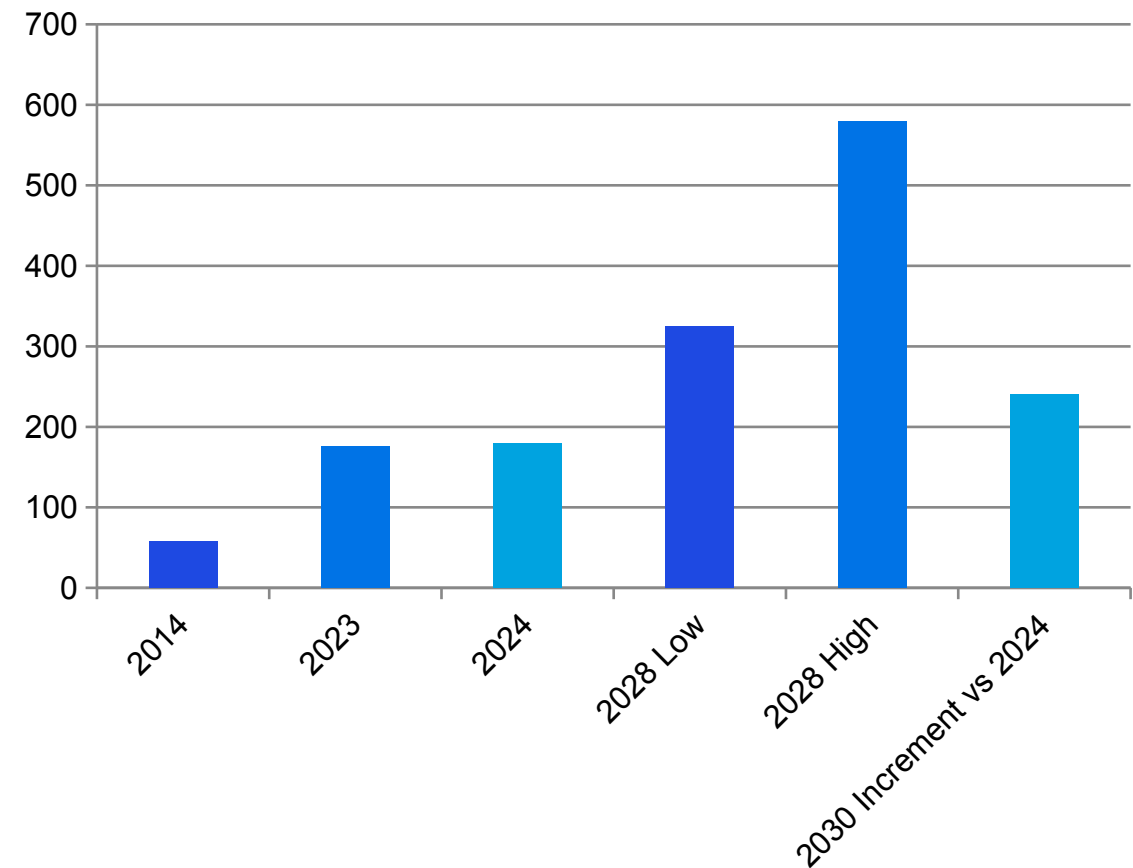


Executive Summary

- U.S. electricity demand growth has re-accelerated, with data centers now a core structural driver rather
- DOE/LBNL estimate data center load rose from 58 TWh (2014) to 176 TWh (2023), with a 2028 range of 325-58
- Data centers represented ~4.4% of U.S. electricity in 2023 and are projected at ~6.7-12% by 2028.
- EIA projects U.S. electricity use growth of 1% in 2026 and 3% in 2027, the strongest 4-year run since 200
- IEA estimates U.S. data center demand around 180 TWh in 2024 and ~+240 TWh by 2030 vs. 2024 levels.
- Grid operators are revising long-term peaks upward; PJM now models materially higher summer and winter pe
- Winning playbooks combine faster interconnection governance, flexible load design, and generation/transmi
- Near-term bottlenecks are less about annual energy and more about coincident peak, queue uncertainty, and
- Leadership teams need a portfolio view: demand quality, locational strategy, and pace-layered supply cont

Demand Growth Curve (2014-2030)

- 2014: 58
- 2023: 176
- 2024: 180
- 2028 Low: 325



U.S. Power System Signals (2024-2027)

Indicator	2024	2025	2026	2027
U.S. electricity demand growth	2.1%	2.3%	2.2%	3.0%*
U.S. data center electricity use (TWh)	~180	n/a	n/a	n/a
U.S. electricity use growth (EIA STEO)	n/a	n/a	1.0%	3.0%
Natural gas generation share (EIA)	n/a	40%	39%	39%
Wind generation share (EIA)	n/a	11%	11%	12%

Regional Reliability Stress: PJM Case

Supply Stack Implications

- Renewables: fast deployment, strong corporate procurement fit, but intermittency requires balancing tools.
- Natural gas: dispatchable bridge for reliability; fuel and permitting risk require portfolio hedging.
- Nuclear/geothermal: strategic long-term firming options, slower commercialization cycles.
- Storage: critical for intraday balancing and peak shaving, less effective for prolonged deficits alone.
- Left side (Speed): renewables + targeted gas + storage provide near-term capacity velocity.
- Right side (Durability): transmission + firm low-carbon assets anchor long-horizon reliability economics.

12-Month Utility-Scale Action Plan

- Month 0-2: Build large-load registry, confidence-score pipeline projects, a
- Month 2-4: Segment projects by energization criticality and locational grid
- Month 4-6: Contract flexible load packages and initiate transmission-ready
- Month 6-9: Lock medium-term supply portfolio and trigger phased interconnec
- Month 9-12: Run reliability drills (summer/winter stress) and re-baseline q
- KPI 1: % of load with validated network deliverability.
- KPI 2: MW of flexible load under executable control.
- KPI 3: Variance between forecast and realized energization dates.

Source Appendix

- DOE press release on 2024 LBNL update (Dec 20, 2024).
- LBNL 2024 U.S. Data Center Energy Usage Report (DOI metadata page).
- EIA press release on Jan 2026 STEO demand update.
- IEA Electricity Mid-Year Update 2025 (demand section).
- IEA Electricity 2025 (demand chapter).
- IEA Energy and AI (energy demand from AI chapter).
- PJM updated 20-year long-term load forecast post.